

Solution Requirements

Date	15 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Patients and doctors must securely log in using email/password with JWT authentication.	High
2	Authorization levels	Role-based access: Patients (book appointments), Doctors (manage schedules), Admins (manage users & reports).	High
3	External interfaces	Integration with Twilio API (notifications), Stripe/PayPal (payments), Google Maps API (clinic location).	High
4	Transactions processing	Appointment booking, payment transactions, and schedule updates must be processed reliably and logged.	High
5	Reporting	Admin dashboard generates reports on appointments, payments, and doctor availability.	Medium
6	Business rules, etc.	Prevent double-booking, enforce cancellation policies, and validate doctor availability before booking.	High
7	Compliance to laws or regulations	Must comply with healthcare data privacy standards (HIPAA-like policies) and secure patient data handling.	High
8	Other	Multi-language support for wider accessibility, telemedicine module for remote consultations.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should respond quickly to user actions.	Page loads in < 2 seconds
2	Scalability	Must handle growth in users and appointments.	Supports up to 10,000 concurrent users
3	Security & Data Privacy	End-to-end encryption, secure API keys, role-based access.	AES-256 encryption, JWT tokens
4	Reliability & Availability	System must be available with minimal downtime.	99.9% uptime per month

5	Usability & Accessibility	User interface must be intuitive and accessible.	WCAG 2.1 AA compliance
6	Maintainability	Codebase should be modular and easy to update.	Clear documentation, reusable modules